Rahul Kumar

CU24250146

BTech CSE 3rd Year SECTION - "C"

Roll No. :- 50

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Experiment No. 1

Experiment Name

Real-World Data Collection, Cleaning and Preprocessing

using Python

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Aim:

To collect a real-world dataset from publicly available sources and perform data preprocessing by handling missing values, duplicate records, inconsistent formats, categorical variables, and outliers using Python.

GITHUB LINK :-https://github.com/DevSharma1912/Mastery-in-Data-Analytics-and-Visualizations-and-Career-Advancement-Technical-/tree/main/Mastery%20in%20Data%20Analytics%20and%20Visualizations%20and%20Career%20Advancement%20(Technical)/Experiment1

Step 1: Import Required Libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from tkinter import Tk, filedialog
from sklearn.preprocessing import LabelEncoder, StandardScaler
```

Step 2: Upload / Select Dataset from Computer

```
import io
from google.colab import files

print("Please select a CSV file to upload.")

uploaded = files.upload()

if not uploaded:
    print("No dataset selected.")
else:
    file_name = next(iter(uploaded)) # Get the name of the first uploaded file
    df = pd.read_csv(io.BytesIO(uploaded[file_name]))

    print("Dataset loaded successfully.")
    print("Selected Dataset:", file_name)
```

```
Please select a CSV file to upload.
Choose Files No file chosen  Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.
Saving WA_Fn-UseC_-HR-Employee-Attrition.csv to WA_Fn-UseC_-HR-Employee-Attrition.csv
Dataset loaded successfully.
Selected Dataset: WA_Fn-UseC_-HR-Employee-Attrition.csv
```

Step 3: Display First Five Records

```
print("\n========== FIRST FIVE RECORDS ==========")

print(df.head())
```

```
========== FIRST FIVE RECORDS ==========
   Age Attrition     BusinessTravel  DailyRate              Department  \
0   41       Yes      Travel_Rarely       1102                   Sales
1   49        No  Travel_Frequently        279  Research & Development
2   37       Yes      Travel_Rarely       1373  Research & Development
3   33        No  Travel_Frequently       1392  Research & Development
4   27        No      Travel_Rarely        591  Research & Development

   DistanceFromHome  Education EducationField  EmployeeCount  EmployeeNumber  \
0                 1          2  Life Sciences              1               1
1                 8          1  Life Sciences              1               2
2                 2          2          Other              1               4
3                 3          4  Life Sciences              1               5
4                 2          1        Medical              1               7

   ...  RelationshipSatisfaction StandardHours  StockOptionLevel  \
0  ...                         1            80                 0
1  ...                         4            80                 1
2  ...                         2            80                 0
3  ...                         3            80                 0
4  ...                         4            80                 1

   TotalWorkingYears  TrainingTimesLastYear WorkLifeBalance  YearsAtCompany  \
0                  8                      0               1               6
1                 10                      3               3              10
2                  7                      3               3               0
3                  8                      3               3               8
4                  6                      3               3               2

  YearsInCurrentRole  YearsSinceLastPromotion  YearsWithCurrManager
0                  4                        0                     5
1                  7                        1                     7
2                  0                        0                     0
3                  7                        3                     0
4                  2                        2                     2
```

```
[5 rows x 35 columns]
```

Step 4: Display Dataset Information

```
print("\n========== DATASET INFORMATION ==========")

print(df.info())
```

```
========== DATASET INFORMATION ==========
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1470 entries, 0 to 1469
Data columns (total 35 columns):
 #   Column                    Non-Null Count  Dtype
---  ------                    --------------  -----
 0   Age                       1470 non-null   int64
 1   Attrition                 1470 non-null   object
 2   BusinessTravel            1470 non-null   object
 3   DailyRate                 1470 non-null   int64
 4   Department                1470 non-null   object
 5   DistanceFromHome          1470 non-null   int64
 6   Education                 1470 non-null   int64
 7   EducationField            1470 non-null   object
 8   EmployeeCount             1470 non-null   int64
 9   EmployeeNumber            1470 non-null   int64
 10  EnvironmentSatisfaction   1470 non-null   int64
 11  Gender                    1470 non-null   object
 12  HourlyRate                1470 non-null   int64
 13  JobInvolvement            1470 non-null   int64
 14  JobLevel                  1470 non-null   int64
 15  JobRole                   1470 non-null   object
 16  JobSatisfaction           1470 non-null   int64
 17  MaritalStatus             1470 non-null   object
 18  MonthlyIncome             1470 non-null   int64
 19  MonthlyRate               1470 non-null   int64
 20  NumCompaniesWorked        1470 non-null   int64
 21  Over18                    1470 non-null   object
 22  OverTime                  1470 non-null   object
 23  PercentSalaryHike         1470 non-null   int64
 24  PerformanceRating         1470 non-null   int64
 25  RelationshipSatisfaction  1470 non-null   int64
 26  StandardHours             1470 non-null   int64
 27  StockOptionLevel          1470 non-null   int64
 28  TotalWorkingYears         1470 non-null   int64
 29  TrainingTimesLastYear     1470 non-null   int64
 30  WorkLifeBalance           1470 non-null   int64
 31  YearsAtCompany            1470 non-null   int64
 32  YearsInCurrentRole        1470 non-null   int64
 33  YearsSinceLastPromotion   1470 non-null   int64
 34  YearsWithCurrManager      1470 non-null   int64
dtypes: int64(26), object(9)
memory usage: 402.1+ KB
None
```

Step 5: Display Statistical Description

```
print("\n========== STATISTICAL DESCRIPTION ==========")

print(df.describe(include="all"))
```

```
========== STATISTICAL DESCRIPTION ==========
                Age Attrition BusinessTravel    DailyRate  \
count   1470.000000      1470           1470  1470.000000
unique          NaN         2              3          NaN
top             NaN        No  Travel_Rarely          NaN
freq            NaN      1233           1043          NaN
mean      36.923810       NaN            NaN   802.485714
std        9.135373       NaN            NaN   403.509100
min       18.000000       NaN            NaN   102.000000
25%       30.000000       NaN            NaN   465.000000
50%       36.000000       NaN            NaN   802.000000
75%       43.000000       NaN            NaN  1157.000000
max       60.000000       NaN            NaN  1499.000000

                  Department  DistanceFromHome    Education EducationField  \
count                   1470       1470.000000  1470.000000           1470
unique                     3               NaN          NaN              6
top     Research & Development               NaN          NaN  Life Sciences
freq                     961               NaN          NaN            606
mean                     NaN          9.192517     2.912925            NaN
std                      NaN          8.106864     1.024165            NaN
min                      NaN          1.000000     1.000000            NaN
25%                      NaN          2.000000     2.000000            NaN
50%                      NaN          7.000000     3.000000            NaN
75%                      NaN         14.000000     4.000000            NaN
max                      NaN         29.000000     5.000000            NaN

       EmployeeCount  EmployeeNumber  ...  RelationshipSatisfaction  \
count         1470.0     1470.000000  ...               1470.000000
unique           NaN             NaN  ...                       NaN
top              NaN             NaN  ...                       NaN
freq             NaN             NaN  ...                       NaN
mean             1.0     1024.865306  ...                  2.712245
std              0.0      602.024335  ...                  1.081209
min              1.0        1.000000  ...                  1.000000
25%              1.0      491.250000  ...                  2.000000
50%              1.0     1020.500000  ...                  3.000000
75%              1.0     1555.750000  ...                  4.000000
max              1.0     2068.000000  ...                  4.000000

       StandardHours  StockOptionLevel  TotalWorkingYears  \
count         1470.0       1470.000000        1470.000000
unique           NaN               NaN                NaN
top              NaN               NaN                NaN
freq             NaN               NaN                NaN
mean            80.0          0.793878          11.279592
std              0.0          0.852077           7.780782
min             80.0          0.000000           0.000000
25%             80.0          0.000000           6.000000
50%             80.0          1.000000          10.000000
75%             80.0          1.000000          15.000000
max             80.0          3.000000          40.000000

       TrainingTimesLastYear  WorkLifeBalance  YearsAtCompany  \
count            1470.000000     1470.000000     1470.000000
unique                   NaN             NaN             NaN
top                      NaN             NaN             NaN
```

Step 6: Check Dataset Shape

```
print("\n========== DATASET SHAPE ==========")

print("Number of Rows:", df.shape[0])
print("Number of Columns:", df.shape[1])
```

```
========== DATASET SHAPE ==========
Number of Rows: 1470
Number of Columns: 35
```

Step 7: Identify Missing Values

```
print("\n========== MISSING VALUES ==========")

missing_values = df.isnull().sum()

print(missing_values)
```

```
========== MISSING VALUES ==========
Age                         0
Attrition                   0
BusinessTravel              0
DailyRate                   0
Department                  0
DistanceFromHome            0
Education                   0
EducationField              0
EmployeeCount               0
EmployeeNumber              0
EnvironmentSatisfaction     0
Gender                      0
HourlyRate                  0
JobInvolvement              0
JobLevel                    0
JobRole                     0
JobSatisfaction             0
MaritalStatus               0
MonthlyIncome               0
MonthlyRate                 0
NumCompaniesWorked          0
Over18                      0
OverTime                    0
PercentSalaryHike           0
PerformanceRating           0
RelationshipSatisfaction    0
StandardHours               0
StockOptionLevel            0
TotalWorkingYears           0
TrainingTimesLastYear       0
WorkLifeBalance             0
YearsAtCompany              0
YearsInCurrentRole          0
YearsSinceLastPromotion     0
YearsWithCurrManager        0
dtype: int64
```

Step 8: Handle Missing Values

```
# Numerical columns -> Median
numerical_columns = df.select_dtypes(
    include=np.number
).columns

for column in numerical_columns:
    if df[column].isnull().sum() > 0:
        df[column] = df[column].fillna(
            df[column].median()
        )

# Categorical columns -> Mode
categorical_columns = df.select_dtypes(
    exclude=np.number
).columns

for column in categorical_columns:
    if df[column].isnull().sum() > 0:
        mode_value = df[column].mode()

        if len(mode_value) > 0:
            df[column] = df[column].fillna(
                mode_value[0]
            )

print("\n========== MISSING VALUE HANDLING ==========")

print("Numerical missing values filled using Median.")
print("Categorical missing values filled using Mode.")

print("\nRemaining Missing Values:")

print(df.isnull().sum().sum())
```

```
========== MISSING VALUE HANDLING ==========
Numerical missing values filled using Median.
Categorical missing values filled using Mode.

Remaining Missing Values:
0
```

Step 9: Detect Duplicate Records

```
duplicate_count = df.duplicated().sum()

print("\n========== DUPLICATE RECORDS ==========")

print("Number of Duplicate Records:", duplicate_count)

# Remove duplicates
df = df.drop_duplicates()

print("Duplicate Records Removed Successfully.")
print("Remaining Rows:", len(df))
```

```
========== DUPLICATE RECORDS ==========
Number of Duplicate Records: 0
Duplicate Records Removed Successfully.
Remaining Rows: 1470
```

Step 10: Standardize Categorical Values

```
print("\n========== CATEGORICAL DATA ==========")

categorical_columns = df.select_dtypes(
    exclude=np.number
).columns

for column in categorical_columns:

    # Remove unnecessary spaces
    df[column] = df[column].astype(str).str.strip()

    # Convert text values to consistent lowercase format
    df[column] = df[column].str.lower()

print("Categorical values standardized successfully.")
```

```
========== CATEGORICAL DATA ==========
Categorical values standardized successfully.
```

Step 11: Encode Categorical Variables

```
encoded_df = df.copy()

categorical_columns = encoded_df.select_dtypes(
    exclude=np.number
).columns

label_encoder = LabelEncoder()

for column in categorical_columns:

    encoded_df[column] = label_encoder.fit_transform(
        encoded_df[column].astype(str)
    )

print("Categorical variables encoded successfully.")
print("Encoding Method: Label Encoding")
```

```
Categorical variables encoded successfully.
Encoding Method: Label Encoding
```

Step 12: Detect Outliers using IQR Method

```
print("\n========== OUTLIER DETECTION ==========")

numeric_columns = encoded_df.select_dtypes(
    include=np.number
).columns

outlier_summary = {}

for column in numeric_columns:

    Q1 = encoded_df[column].quantile(0.25)
    Q3 = encoded_df[column].quantile(0.75)

    IQR = Q3 - Q1

    lower_bound = Q1 - 1.5 * IQR
    upper_bound = Q3 + 1.5 * IQR

    outliers = encoded_df[
        (encoded_df[column] < lower_bound) |
        (encoded_df[column] > upper_bound)
    ]

    outlier_summary[column] = len(outliers)

    print(
        column,
        "-> Outliers:",
        len(outliers)
    )
```

```
========== OUTLIER DETECTION ==========
Age -> Outliers: 0
Attrition -> Outliers: 237
BusinessTravel -> Outliers: 0
DailyRate -> Outliers: 0
Department -> Outliers: 0
DistanceFromHome -> Outliers: 0
Education -> Outliers: 0
EducationField -> Outliers: 0
EmployeeCount -> Outliers: 0
EmployeeNumber -> Outliers: 0
EnvironmentSatisfaction -> Outliers: 0
Gender -> Outliers: 0
HourlyRate -> Outliers: 0
JobInvolvement -> Outliers: 0
JobLevel -> Outliers: 0
JobRole -> Outliers: 0
JobSatisfaction -> Outliers: 0
MaritalStatus -> Outliers: 0
MonthlyIncome -> Outliers: 114
MonthlyRate -> Outliers: 0
NumCompaniesWorked -> Outliers: 52
Over18 -> Outliers: 0
OverTime -> Outliers: 0
PercentSalaryHike -> Outliers: 0
PerformanceRating -> Outliers: 226
RelationshipSatisfaction -> Outliers: 0
StandardHours -> Outliers: 0
StockOptionLevel -> Outliers: 85
TotalWorkingYears -> Outliers: 63
TrainingTimesLastYear -> Outliers: 238
WorkLifeBalance -> Outliers: 0
YearsAtCompany -> Outliers: 104
YearsInCurrentRole -> Outliers: 21
YearsSinceLastPromotion -> Outliers: 107
YearsWithCurrManager -> Outliers: 14
```

Step 13: Visualize Outliers using Box Plots

```
boxplot_columns = encoded_df.select_dtypes(
    include=np.number
).columns

if len(boxplot_columns) > 0:

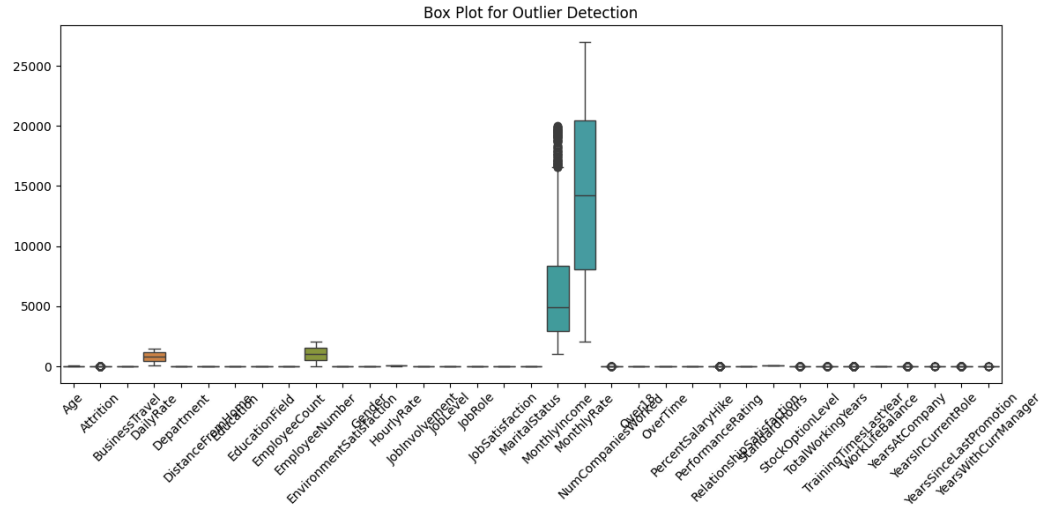
    plt.figure(figsize=(12, 6))

    sns.boxplot(
        data=encoded_df[boxplot_columns]
    )

    plt.title("Box Plot for Outlier Detection")
    plt.xticks(rotation=45)
    plt.tight_layout()

    plt.show()

print("Box Plot generated successfully.")
```




```
Box Plot generated successfully.
```

Step 14: Treat Outliers using IQR Capping

```
for column in numeric_columns:

    Q1 = encoded_df[column].quantile(0.25)
    Q3 = encoded_df[column].quantile(0.75)

    IQR = Q3 - Q1

    lower_bound = Q1 - 1.5 * IQR
    upper_bound = Q3 + 1.5 * IQR

    encoded_df[column] = encoded_df[column].clip(
        lower_bound,
        upper_bound
    )

print("Outliers treated using IQR Capping.")
```

```
Outliers treated using IQR Capping.
```

Step 15: Normalize / Standardize Numerical Attributes

```
numeric_columns = encoded_df.select_dtypes(
    include=np.number
).columns

scaler = StandardScaler()

if len(numeric_columns) > 0:

    encoded_df[numeric_columns] = scaler.fit_transform(
        encoded_df[numeric_columns]
    )

print("\n========== FEATURE SCALING ==========")

print("Numerical attributes standardized successfully.")
print("Scaling Method: Standardization")
print("Mean approximately equals 0.")
print("Standard deviation approximately equals 1.")
```

```
========== FEATURE SCALING ==========
Numerical attributes standardized successfully.
Scaling Method: Standardization
Mean approximately equals 0.
Standard deviation approximately equals 1.
```

Step 16: Feature Engineering

```
if "sibsp" in encoded_df.columns and "parch" in encoded_df.columns:

    encoded_df["family_size"] = (
        encoded_df["sibsp"] +
        encoded_df["parch"] +
        1
```

```
    )

    print("\nFeature Created: family_size")


# If Age is available, create AgeGroup
if "age" in df.columns:

    age_values = df["age"]

    encoded_df["age_group"] = pd.cut(
        age_values,
        bins=[0, 12, 18, 35, 60, 100],
        labels=[
            "child",
            "teenager",
            "adult",
            "middle_aged",
            "senior"
        ]
    )

    # Encode Age Group
    encoded_df["age_group"] = LabelEncoder().fit_transform(
        encoded_df["age_group"].astype(str)
    )

    print("Feature Created: age_group")
```

Step 17: Display Preprocessed Dataset

```
print("\n========== PREPROCESSED DATASET ==========")

print(encoded_df.head())
```

```
========== PREPROCESSED DATASET ==========
        Age  Attrition  BusinessTravel  DailyRate  Department  \
0  0.446350        0.0        0.590048   0.742527    1.401512
1  1.322365        0.0       -0.913194  -1.297775   -0.493817
2  0.008343        0.0        0.590048   1.414363   -0.493817
3 -0.429664        0.0       -0.913194   1.461466   -0.493817
4 -1.086676        0.0        0.590048  -0.524295   -0.493817

   DistanceFromHome  Education  EducationField  EmployeeCount  EmployeeNumber  \
0         -1.010909  -0.891688       -0.937414            0.0       -1.701283
1         -0.147150  -1.868426       -0.937414            0.0       -1.699621
2         -0.887515  -0.891688        1.316673            0.0       -1.696298
3         -0.764121   1.061787       -0.937414            0.0       -1.694636
4         -0.887515  -1.868426        0.565311            0.0       -1.691313

   ...  RelationshipSatisfaction  StandardHours  StockOptionLevel  \
0  ...                 -1.584178            0.0         -0.978160
1  ...                  1.191438            0.0          0.300537
2  ...                 -0.658973            0.0         -0.978160
3  ...                  0.266233            0.0         -0.978160
4  ...                  1.191438            0.0          0.300537

   TotalWorkingYears  TrainingTimesLastYear  WorkLifeBalance  YearsAtCompany  \
0          -0.424999              -2.130765        -2.493820       -0.123832
1          -0.151076               0.278639         0.338096        0.678078
2          -0.561960               0.278639         0.338096       -1.326697
3          -0.424999               0.278639         0.338096        0.277123
4          -0.698921               0.278639         0.338096       -0.925742

   YearsInCurrentRole  YearsSinceLastPromotion  YearsWithCurrManager
0           -0.058474                -0.776139              0.253823
1            0.785616                -0.372486              0.822605
2           -1.183926                -0.776139             -1.168130
3            0.785616                 0.434819             -1.168130
4           -0.621200                 0.031166             -0.599348

[5 rows x 35 columns]
```

Step 18: Display Final Dataset Information

```
print("\n========== FINAL DATASET INFORMATION ==========")

print("Rows:", encoded_df.shape[0])
print("Columns:", encoded_df.shape[1])

print("\nMissing Values:")
print(encoded_df.isnull().sum().sum())

print("\nDuplicate Records:")
print(encoded_df.duplicated().sum())
```

```
========== FINAL DATASET INFORMATION ==========
Rows: 1470
Columns: 35

Missing Values:
0

Duplicate Records:
0
```

Step 19: Save Cleaned Dataset

```
output_file = "cleaned_preprocessed_dataset.csv"

encoded_df.to_csv(
    output_file,
    index=False
)

print("\n========== DATASET STORAGE ==========")

print(
    "Cleaned Dataset Saved Successfully:",
    output_file
```

```
)
```

```
========== DATASET STORAGE ==========
Cleaned Dataset Saved Successfully: cleaned_preprocessed_dataset.csv
```

Step 20: Final Output

```
print("\n========== DATA PREPROCESSING ==========")

print("Dataset Collection: Completed")
print("Dataset Loading: Completed")
print("Data Inspection: Completed")
print("Missing Value Handling: Completed")
print("Duplicate Removal: Completed")
print("Categorical Standardization: Completed")
print("Categorical Encoding: Completed")
print("Outlier Detection: Completed")
print("Outlier Treatment: Completed")
print("Feature Standardization: Completed")
print("Feature Engineering: Completed")
print("Cleaned Dataset Export: Completed")

print("\nObservations:")

print("Missing numerical values were handled using the median.")
print("Missing categorical values were handled using the mode.")
print("Duplicate records were removed.")
print("Categorical values were standardized and encoded.")
print("Outliers were detected using the IQR method.")
print("Outliers were treated using IQR capping.")
print("Numerical features were standardized using StandardScaler.")
print("New features were created where applicable.")

print("\nExperiment Completed Successfully.")
```

```
========== DATA PREPROCESSING ==========
Dataset Collection: Completed
Dataset Loading: Completed
Data Inspection: Completed
Missing Value Handling: Completed
Duplicate Removal: Completed
Categorical Standardization: Completed
Categorical Encoding: Completed
Outlier Detection: Completed
Outlier Treatment: Completed
Feature Standardization: Completed
Feature Engineering: Completed
Cleaned Dataset Export: Completed

Observations:
Missing numerical values were handled using the median.
Missing categorical values were handled using the mode.
Duplicate records were removed.
Categorical values were standardized and encoded.
Outliers were detected using the IQR method.
Outliers were treated using IQR capping.
Numerical features were standardized using StandardScaler.
New features were created where applicable.

Experiment Completed Successfully.
```

EXPERIMENT 1 - ANSWERS TO QUESTIONS

QUESTION 1: Why is data preprocessing considered one of the most important phases in data analytics?

Data preprocessing is important because real-world data is often incomplete, inconsistent, duplicated and noisy.

Before performing visualization, statistical analysis or machine learning, the data must be cleaned and prepared.

Data preprocessing helps to:

Handle missing values. Remove duplicate records. Correct inconsistent data. Detect and treat outliers. Convert categorical variables into numerical form. Bring numerical features to a common scale.

Proper preprocessing improves the quality and reliability of analytical results.

QUESTION 2: Explain different methods of handling missing values with suitable examples.

Missing values can be handled using several methods.

Mean: Missing numerical values can be replaced by the mean of the available values.

Median: Missing numerical values can be replaced by the median. Median is useful when the data contains outliers.

Mode: Missing categorical values can be replaced by the most frequently occurring value.

Row Removal: Rows containing missing values can be removed when the number of missing records is small.

Column Removal: A column can be removed when a very large portion of its values is missing.

Example:

If the Age column contains missing values:

df["Age"] = df["Age"].fillna(df["Age"].median())

This replaces missing Age values with the median age.

QUESTION 3: Differentiate between Label Encoding and One-Hot Encoding.

Label Encoding:

Label Encoding converts categories into numerical labels.

Example:

Male → 0 Female → 1

It is useful when categories have an inherent order or when a compact numerical representation is required.

One-Hot Encoding:

One-Hot Encoding creates separate binary columns for each category.

Example:

Gender

Male Female

becomes:

Male Female 1 0 0 1

Main Difference:

Label Encoding → One numerical value per category. One-Hot Encoding → Separate binary column for each category.

QUESTION 4: What are outliers? How can they affect analytical results?

Outliers are observations that are significantly different from the majority of values in a dataset.

They can occur because of:

Measurement errors. Data entry errors. Unusual events. Natural variation.

Outliers can affect:

Mean and standard deviation. Statistical analysis. Correlation. Machine learning models. Data visualization.

The IQR method can be used to identify outliers.

IQR = Q3 - Q1

Lower Bound = Q1 - 1.5 × IQR

Upper Bound = Q3 + 1.5 × IQR

QUESTION 5: Explain the difference between normalization and standardization.

Normalization:

Normalization generally scales values to a fixed range, commonly 0 to 1.

A common formula is:

X_normalized = (X - X_min) / (X_max - X_min)

Standardization:

Standardization transforms values so that the feature has approximately mean 0 and standard deviation 1.

Formula:

X_standardized = (X - Mean) / Standard Deviation

Main Difference:

Normalization → Fixed range, commonly 0 to 1. Standardization → Mean approximately 0 and standard deviation approximately 1.

QUESTION 6: Why should duplicate records be removed before analysis?

Duplicate records represent repeated observations of the same data.

If duplicates are not removed, they can:

Increase the apparent frequency of certain observations. Produce biased statistical results. Affect averages and distributions. Affect machine learning model training. Increase the apparent size of the dataset.

Removing duplicate records helps ensure that each observation is represented appropriately in the dataset.

QUESTION 7: What is feature engineering? Give two practical examples.

Feature engineering is the process of creating new useful attributes from existing data.

It can help represent information in a form that is more useful for analysis or machine learning.

Example 1:

Family Size can be created from:

SibSp + Parch + 1

FamilySize = SibSp + Parch + 1

Example 2:

Age Group can be created from Age.

For example:

0-12 → Child 13-18 → Teenager 19-35 → Adult 36-60 → Middle Aged 60+ → Senior

Feature engineering can provide additional useful information without collecting new raw data.

QUESTION 8: Which preprocessing techniques would you apply to the IBM HR Employee Attrition dataset and why?

For the IBM HR Employee Attrition dataset, the following preprocessing techniques can be applied:

Missing Value Handling: Identify and appropriately handle missing values.

Duplicate Removal: Remove duplicate employee records.

Categorical Encoding: Convert categorical attributes such as department, job role and marital status into numerical form.

Outlier Detection: Use box plots and the IQR method for numerical variables.

Feature Scaling: Standardize numerical features such as age, income and years of experience where appropriate.

Feature Engineering: Create useful attributes such as age groups or income categories.

These techniques improve data quality and prepare the employee dataset for statistical analysis and machine learning.

QUESTION 9: How does poor-quality data affect machine learning model performance?

Poor-quality data can negatively affect machine learning models.

Problems such as:

Missing values. Duplicate records. Incorrect values. Outliers. Inconsistent categories. Different numerical scales.

can cause models to learn incorrect or misleading patterns.

This may result in:

Lower prediction accuracy. Poor generalization. Biased predictions. Unstable models. Reduced reliability.

Therefore, high-quality preprocessing is important for building reliable machine learning models.

QUESTION 10: Name any three Python libraries commonly used for data preprocessing.

Three commonly used Python libraries are:

Pandas: Used for loading, cleaning, manipulating and analyzing structured datasets.

NumPy: Used for numerical operations and array processing.

Scikit-learn: Provides preprocessing tools such as LabelEncoder, StandardScaler and other machine learning utilities.

Other useful libraries include Matplotlib and Seaborn for visualization and exploratory data analysis.